COMP 100: Introduction to Computer Science and Programming
Fall 2024

Course Description

This course is intended for Computer Engineering students in their first year with **little or no programming experience**. It is also suitable for students from any other department who simply want to learn how to program because it's a very useful skill. It aims to provide students with an understanding of the role computation can play in solving problems. By the end of the course a student

1. Will be comfortable coming up with a procedural solution to a problem and coding it in Python.
2. Will be able to apply fundamental concepts in Computer Science such as decomposition, and abstraction.

Course Overview

Lectures will cover fundamental concepts illustrated with examples and with in-class exercises. Prelabs and problem sets will delve deeply into all the topics covered by the course. Friday lab sessions involve hands-on programming drills, with active help from TAs, who will go over the solutions and address any conceptual difficulties.

A significant portion of the material for this course will be presented only in the lecture, so students are expected to regularly attend lectures. Attending Labs is **mandatory.**

The course is fast-paced, so please put in the work to keep up and avoid falling behind! Apart from the above opportunities to learn, the instructor and TAs will have office hours every week to address doubts and questions about the material. No prior appointment is necessary for office hours; just show up.

Instructor: Sajit Rao

KU Credits: 3

Language of Instruction: English

Lectures: 2 Lectures every week**.** There are 3 sections

Office Hours: TBA

Teaching Assistants (TAs): TBA

Catalog Description: Overview of Computers, Programming, Algorithms, and Programming Languages. Programming with Python: Data types, variables, operators. Control statements: conditionals, loops, iteration. String manipulation. Functions, recursion, decomposition and abstraction. Tuples, lists, dictionaries. Aliasing, mutability, cloning. Files. Object oriented programming, classes, inheritance. Testing, debugging, exception handling.

Course Textbook

The textbook is: Guttag, John. *Introduction to Computation and Programming Using Python: With Application to Understanding Data Second Edition.* MIT Press, 2016. ISBN: 9780262529624. There is more detail in the book about some of the lecture topics. It is available both in hard copy and as an ebook.

Course Learning Outcomes (CLOs)

1) Implement a given algorithm as a computer program in Python.
2) Explain what a given program in Python does and learn and practice with testing and debugging techniques to identify errors in a program.
3) Understand and follow good programming practices in coding such as Abstraction and Decomposition via functions.
4) Learn about variable scopes in Python.
5) Learn control flow via branching and different types of loops.
6) Learn and use different data types in Python such as strings, lists, tuples, dictionaries.
7) Learn the basics of Error Handling.
8) Learn recursive thinking and adapt it to different problems.
9) Learn basic principles of Object-Oriented Programming and Inheritance in Python.
10) Learn the basics of bash shell and simple commands to interact with the Operating System.
11) Learn the basics of program efficiency.

Assessment Methods

Method	**Weight**
Problem Sets & Labs	20%
In-class Exercises/Pop-quizzes	5%
Midterm	35%
Final	40%

All percentages are subject to change within a margin of 5% during the semester at the instructor's discretion.

Problem Sets: The problem sets will be due anywhere from one to two weeks from the moment they are issued (depending on how long they are). **You must complete all problem sets to pass the course.**

Attendance: You must physically attend **at least 70% of the labs** to pass the course. **If your lab attendance is below 70%, you will not be eligible to take the final**, i.e. you will get an F.

Lab Recitations: Lab-Recitations every week will consist of solving some pre-lab exercises before the lab, some in-lab exercises during the lab, and going over their solutions and any conceptual difficulties with the TAs. **Attending Labs is mandatory (attendance will be taken).**

Exams: The Midterm and Final exam will be conducted face-to-face. The average of the midterm and final scores must be **at least 33%** to pass the course.

A passing grade may be converted to a failing grade under *any* of the following circumstances:

1. If you do not turn in **All** the problem sets in time, where "turn in" means making a serious attempt to complete each problem by yourself.

2. If you **cheat.** For example, copy another person's work, or use online help or A.I agents (such as, but not restricted to ChatGPT) and represent it as your own work **in any single** problem set, lab or exam.

3. If your average score from the midterm and Final exams is **below 33%**

4. You attend **less than 70%** of the Labs

5. Your overall score (out of 100) is less than some to-be-determined X. Where X is chosen based on the whole class performance at the end of the semester (it is typically between 30-40%)

List of Topics

- Introduction to computers and computation
- Objects, types, variables, and simple operations
- Functions, Decomposition, Abstraction
- Branching and iteration
- Guess and check, approximations, bisection
- Strings and string manipulation,
- Tuples, Lists,
- Aliasing, mutability, cloning, nested data structures

- Recursion
- Dictionaries,
- File I/O
- Introduction to object-oriented programming, classes and inheritance
- Shell and shell tools
- Testing, debugging, exceptions, assertions
- Understanding program efficiency
- [tentative] Modules: Numpy, Pandas, Matplotlib

Late Submission Policy

Problem sets submitted late (i.e. after the deadline) incur the following penalties

- Late by 10 hrs or less: **20% penalty**
- More than 10 hrs: **submission is not accepted**.

Please start your assignments early and do not ask for extensions.

Prelabs must be submitted at the latest **by the published deadline** (usually before the start of the first lab on Fridays), any submission after that will not be accepted.

Labs must be submitted **by the end of the lab session**, submissions after that are not accepted.

Exam and Make-Up Policy

There will be one midterm, and one final exam.

If a student misses an exam with a valid excuse, he/she can apply for a make-up exam. For excuses to be valid, they must be accepted by Koç University and communicated with the instructor through official channels. Emergencies must be properly documented, and medical reports must be approved by Koç University Health Center.

A **single make-up exam** for the midterm will be given within 10 days of the original midterm.

A **single make-up exam** for the final exam will be scheduled by the registrar during the make-up exam period after finals.

Students who have **not met the minimum lab attendance requirements** (see above) **will not be allowed to take the final exam**, and automatically fail the course.

There will be **no further make-up exams** for those who miss a make-up exam.

As this course is offered both in the Fall and Spring semesters, **there will be no Remedial exam** offered to those who fail the course.

All exams are planned to be conducted physically, in-person. **We will not offer online exams and/or alternative means of evaluation under any circumstances.**

Academic Honesty Policy

Honesty and trust are important to all of us as individuals. Students and faculty adhere to the following principles of academic honesty at Koç University:

- Individual accountability for all individual work, written or oral. Copying from others or providing answers or information to others, written or oral, is cheating.
- Providing proper acknowledgement of original author. Copying from another student's paper or from another source without written acknowledgement is plagiarism.
- Getting unauthorized help from another person or having someone else write one's paper or assignment is collusion.

Academic dishonesty in the form of cheating, plagiarism, or collusion are serious offenses and are not tolerated at Koç University. By taking this course, students acknowledge that they must fully comply with Koç University's Student Code of Conduct (https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/). University Academic Regulations and the Regulations for Student Disciplinary Matters clearly define the policy and the disciplinary action to be taken in case of academic dishonesty. Failure in academic integrity may lead to suspension and expulsion from the University.

All work in the graded components of the course must belong to the student. We would like to remind all students that the following actions will be treated as violations of academic honesty and met with disciplinary action:

- Working jointly with another student on a problem set, lab or exam
- Sharing problem set, lab or exam answers with others
- Using AI tools like ChatGPT to answer questions on problem sets, labs, or exams
- Looking at another student's problem set, lab or exam
- Having someone else do a problem set, lab or exam for you (paid or not)
- Making copies of lab, problem set or exam questions and distributing them to others